

13.2 AI-Enabled End-to-End Design in RFICs with Controllable Architectural Style from ‘Classical’ to ‘Non-Intuitive’ for mm-Wave/sub-THz LNAs

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Abstract

This paper introduces a unified algorithmic design flow for low-noise amplifiers (LNAs), spanning specifications to layout and integrating topology, architecture, circuit, and electromagnetic (EM) design in one framework. The algorithm supports designer input from

classical to unconventional architectures, enhancing interpretability, debugging, and usability. We demonstrate two LNAs covering 24 to 150GHz with state-of-the-art performance, including novel architectures uncovered by the algorithm.

This paper introduces a unified algorithmic design flow for low-noise amplifiers (LNAs) that spans from specifications to layout, integrating topology, architecture, circuit, and electromagnetic (EM) design in an end-to-end framework. The approach incorporates controllable EM interfaces ranging from classical transmission-line shapes to arbitrary pixelated structures, each offering distinct design trade-offs. Unlike prior AI-enabled inverse design methods for EM and passive synthesis [1-3], this framework supports controlled architectural evolution, enabling designer preference, interpretability, and seamless integration with surrounding circuits—facilitating debugging, fault isolation, and broader usability. In addition, this paper also demonstrates unique topology choices for the LNA that emerge as a result of the algorithmic approach that would not have been possible with a classical designer-driven synthesis.

Designing LNAs to meet targets for gain, matching, noise figure, stability, linearity, and power efficiency requires multi-dimensional optimization across architecture, topology, stage count, and device type (e.g., common-emitter/source or common-base/gate), along with EM co-design using lumped or distributed elements. Conventional RFIC design depends on fixed templates and iterative tuning; if unmet, new architectures are introduced, followed by circuit and EM optimization at layout to handle parasitics and coupling. This process is lengthy—often months—and constrained by designer intuition regarding feasible topologies and EM structures

Here, we propose an end-to-end parasitic and EM-aware algorithmic design flow for RF/mmWave ICs, particularly focusing on LNA design that enables specifications to layout synthesis, allowing: 1) discovery of novel LNA architectures that are extremely challenging to achieve with human designs due to the lack of suitable templates, 2) topology and architecture selection, 3) LNA circuit and parameter optimization, 4) full circuit-EM co-design and synthesis covering both classical and non-intuitive structures, and 5) end-to-end optimization allowing specifications to layout-ready synthesis. We demonstrate the methodology with proof-of-concept LNAs spanning 30 to 160GHz. The first example is a frequency-diplexing LNA achieving near-optimal performance covering all mmWave frequency bands from 24-to-90 GHz, while overcoming the trade-offs associated with bandwidth, efficiency and noise, by employing highly efficient non-intuitive arbitrary-shaped three-port broadband frequency-diplexing matching networks. The LNA achieves an effective bandwidth of 24-to-90GHz with peak gain of 16dB and noise factor (NF) ranging from 3.8 to 5.7dB. It can be switched between a 24-to-55GHz low-frequency mode, a 55-to-90GHz high-frequency mode, or a combined path mode. We also present a fully AI-synthesized LNA using classical transmission-line-based matching networks exhibiting dual resonances with competitive bandwidth, noise figure, and gain, demonstrating peak gains of 30dB and 18dB at 85GHz and 150GHz respectively. To our knowledge, this is the first demonstration of a universal algorithmic flow capable of synthesizing RFICs from specifications to layout, validated through measurement of both classical and non-intuitive topologies.

The overarching vision of this work is summarized in Fig. 13.2.1. Mapping from an RFIC to its specifications is unique, whereas the reverse mapping—from specifications to a realizable RFIC (architecture, topology, circuit, EM, and parameters)—is inherently one-to-many. Multiple designs can satisfy the same specifications, with final choices often shaped by subjective considerations such as explainability, simplicity, area, sensitivity, debugging, and interfacing with ancillary circuits. Thus, controlling the evolution of architecture is essential for the broader adoption of AI-enabled RFIC design. Figure 13.2.2 outlines the proposed flow for LNA design. Specifications and architectural style (classical transmission-line vs. pixelated) serve as inputs. For pixelated architectures, synthesis proceeds in two stages: (1) reinforcement learning (RL) explores optimal architectures, including topology, stage configuration, device sizing, biasing, and interface impedances to meet specifications and (2) inverse EM design generates structures that satisfy target scattering parameters in the chosen “style”. We demonstrate two strategies for pixelated synthesis: one employs a forward AI model with metaheuristics to generate EM structures, while the other incorporates transmission-line parameters directly into the RL process, enabling joint circuit-EM optimization. Through this, the designer can enforce their own constraints on topology, architecture, and interpretability, where prior inverse works did not allow for EM structure “type” selection between pixelated or classical [4-6].

Figure 13.2.2 depicts the integration of reinforcement learning (RL) to optimize circuit architecture, topology, parameters, and impedance matching, with the latter subsequently

dispatched to the inverse algorithm for optimization and actual structure synthesis. The encoded design space accommodates architectural options (e.g., number of stages), topological variations (e.g., common-emitter, common-base, cascode), device-level parameters (e.g., transistor sizing, biasing), and interface impedances, while also incorporating operating classes and layout-aware constraints for joint circuit and electromagnetic (EM) synthesis. As shown in Figure 13.2.2, the RL engine is formulated as a Markov decision process where the policy network maps design states and performance metrics to a probability distribution over actions, guiding optimization toward multi-objective specifications. The reward function combines hard and soft targets, penalizing deviations from constraints. Circuit parameters are linked with an AI-assisted inverse network that generates compact EM structures consistent with desired S-parameters [7], thereby extending the design space beyond conventional heuristics. Training with ~350,000 circuit-EM examples requires ~24 hours on a 192-core cluster using proximal policy optimization (PPO) [8] with a three-layer multilayer perceptron (MLP) actor of 128 neurons per layer and a three-layer MLP critic of 256 neurons per layer. The training data is generated directly from the model’s interactions with a commercial circuit simulator and a custom EM surrogate model, resulting in high accuracy. Data to train the EM surrogate model comes from simulating random examples within the design space in a commercial EM simulator. Once trained, output-stage layouts are synthesized in 5–10 minutes, while a complete low-noise amplifier (LNA) design, including auxiliary circuitry, is achieved within hours. Partitioning circuit and EM synthesis enables efficient end-to-end optimization. It is important to note that the architectural choice between classical and non-intuitive does not come at a cost performance. As can be seen in Fig. 13.2.3, different EM structures (both classical and non-classical) can achieve similar EM performance that can be guided through an algorithm. Compared to [9] for PCB based filter designs, this work avoids the training of any diffusion models, and demonstrates on-chip realization of mmWave/sub-THz passives that are integrated with the circuit synthesis in a tight loop.

AI not only allows synthesis of particular circuits, but also allows canvassing of possible designs in a rapid fashion, allowing us to evaluate the Pareto front achievable in a given process design kit. This is often an overlooked benefit. Figure 13.2.4 demonstrates the RL-based synthesis of LNAs with the proposed methods demonstrating achieved target gains and NF with the minimum power dissipation. It is to be noted that due to the parasitic and EM-aware training process, the reported numbers are the final end-to-end simulated RF performance for the GDS-ready circuit. As can be seen, the simulated performances are competitive with state of the art, demonstrating that the RL algorithm has been able to learn the design space effectively. The example designs in the figure demonstrate that the algorithm can effectively choose circuit topologies, bias them properly for optimal noise performance, add gain stages when needed, and synthesize the optimal impedance as well. More importantly, as shown in the figure, the design allows us to quantify the Pareto front of the design space. The figure shows the trade-off of NF and gain with DC power, and other considerations such as linearity, area, robustness, sensitivity, stability measures can easily be added, allowing a designer to evaluate the PDK and get to the optimal design orders of magnitudes faster.

The paper presents two design examples of LNAs demonstrating the pixelated and classical approaches, particularly focusing on how algorithms can allow synthesis of architectures that would be very challenging for a human designer. The first LNA is designed to cover the mmWave spectrum from 24 to 90GHz for joint sensing and communication, covering 5G mmWave, satellite communication, and radar as a broadband front-end. Broadband design presents key challenges, particularly achieving simultaneous power and noise matching across wide frequencies. Matching networks must provide flat gain while incurring minimal loss and noise penalties. Leveraging arbitrary multi-port synthesis, we introduce a frequency-diplexing LNA that separates matching into two bands: 24 to 55GHz and 55 to 90GHz. The two-stage design employs cascode and common-emitter stages, with input/output networks enabling diplexing. The input directs the 24-to-55GHz band through the upper branch for optimal power/noise match without leakage, while the 55-to-90GHz band is directed oppositely; the output combiner mirrors this behavior. Designing such low-loss broadband networks is difficult due to absent templates, but the algorithmic flow enables their realization. As shown in the measurement results in Fig. 13.2.6, the LNA effectively realizes 16dB gain across a record bandwidth of 24 to 90GHz, with state of the art NF (3.8 to 6.8dB) across the entire spectrum matching very closely with simulations,

dissipating approximately 24mW of power in the low-band mode and 32mW of power in the high-band mode.

To demonstrate the synthesis with classical t-lines, we present a dual peaking LNA between 85GHz and 150GHz allowing for dual purpose usage at radar and 6G frequencies. Designing multi-peaking requires careful tuning to align the matching while providing simultaneous noise-optimal matching, all of which are taken into account during the RL training. The measurement results are seen in Fig. 13.2.6, where the dual peaking LNA realizes a peak gain of 30dB and minimum NF of 5.8dB at 85GHz and a peak gain of 18dB at 160GHz, with a power dissipation of 63mA.

A table of comparison with state of the art is shown Fig. 13.2.7. The presented LNAs demonstrate state of the art performance and are synthesized for the first time through an end-to-end AI algorithm that allows controllable architectural choice as designer inputs for broad applicability.

Acknowledgement:

The authors would like to acknowledge GlobalFoundries for chip fabrication support, and the funding agencies: Army Research Office (W911N1NF2110314, W911N1NF2401011), Air Force Office of Scientific Research (FA9550-23-1-0176), Office of Naval Research (N00014-23-1-2592), DURIP funding (N00014-23-1-2332), and National Science Foundation (CNS-2211617, CNS-2148271, CNS-2402782). Authors also would like to thank Bert Harrop from the Princeton University Physics Department for professional assistance with packaging, and Princeton ECE IMRL for design support and discussions.

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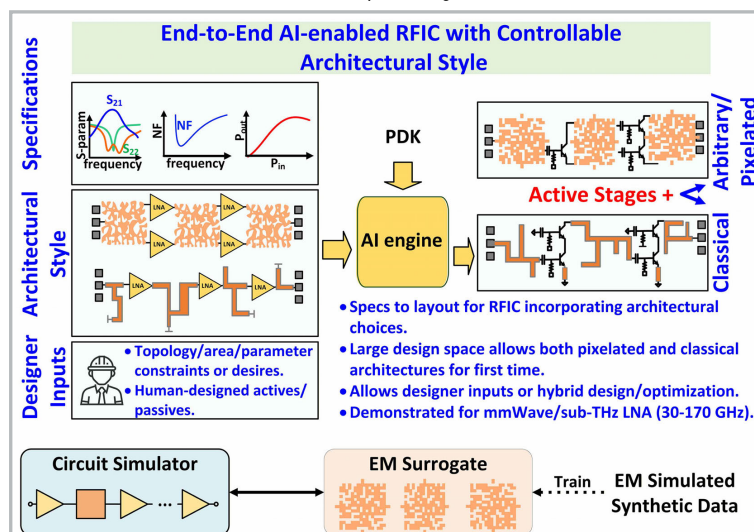


Figure 13.2.1: RFIC synthesis from specifications to layout allowing designer choice of either classical or non-intuitive pixelated architecture for interpretability and debugging. A commercial circuit simulator and EM surrogate are used for accurate results.

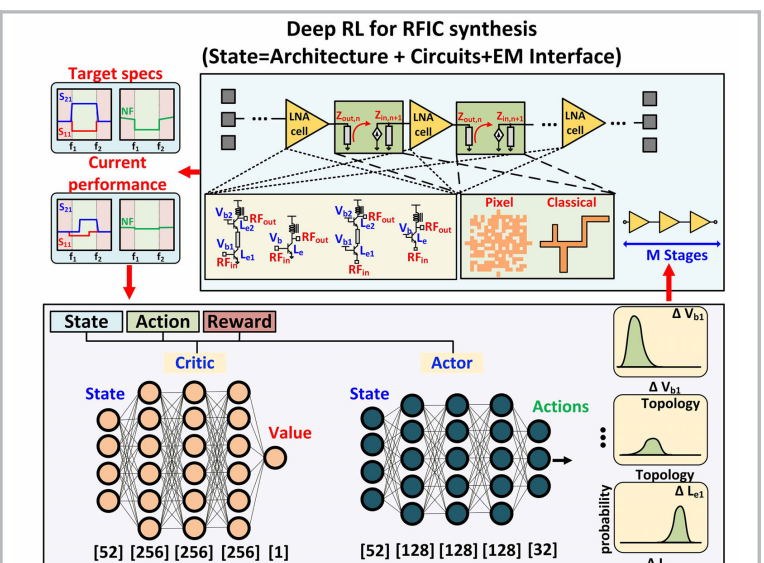


Figure 13.2.2: Deep reinforcement learning algorithm to allow optimization over architecture, circuit topology, parameters and full EM-circuit co-design incorporating classical t-line optimization and pixelated architecture.

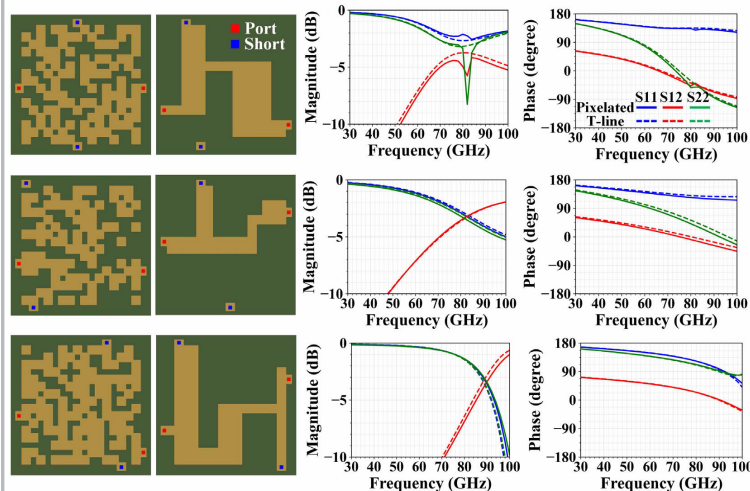


Figure 13.2.3: The same EM performance can be realized by either inverse designed pixelated structures or more 'classical' transmission-line based structures as demonstrated by the example in 90nm SiGe BiCMOS showing near-identical performance across all scattering parameters.

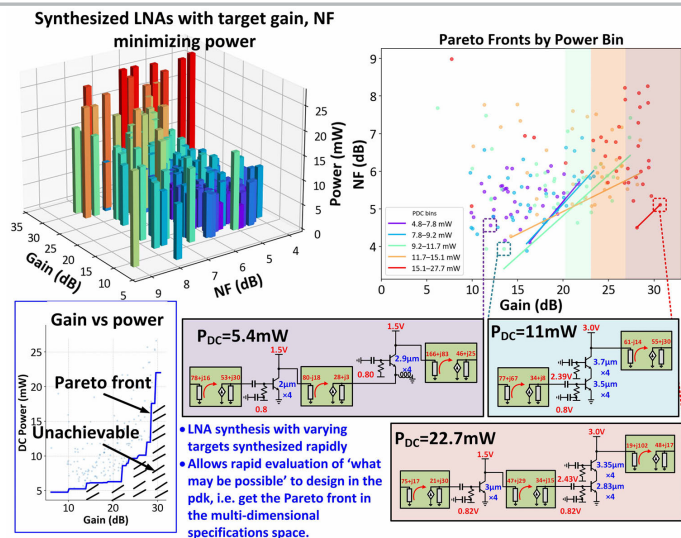


Figure 13.2.4: Algorithmic synthesis approaches can demonstrate the 'Pareto fronts' for RFICs indicating what might be possible to synthesize in the process design kit, as shown in the synthesized examples of LNA achieving varying gain and NF, while minimizing power consumption.

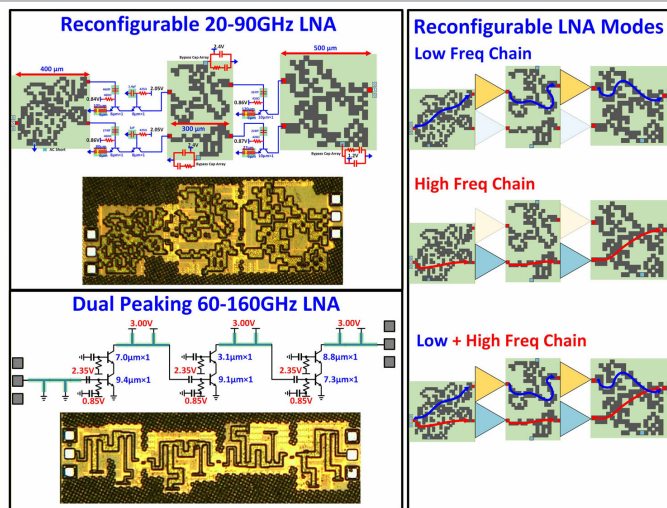


Figure 13.2.5: Schematic and chip micrographs of the two proof-of-concept LNAs, with the first example demonstrating a frequency duplexing architecture (enabling NF minimization) covering all mmWave bands between 20-to-90GHz aided by an inverse designed 3-port structure, and the second demonstrating a dual band LNA with a more classical architecture with peaks at 85 and 150 GHz.

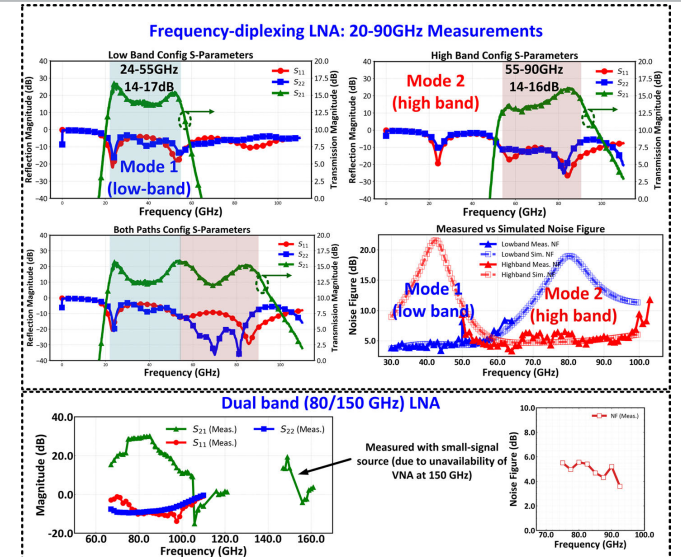


Figure 13.2.6: Measurement results demonstrating the switched-mode frequency duplexing operating from 20-to-90GHz with state-of-the-art NF, and measurements of the dual band LNA.

Reference	This work: 24-90GHz Reconfigurable LNA	[10] Park, ISSCC'22	[11] Zhao, JSSC'25	[12] Mendes, MTT'25	[13] Yu, JSSC'17	[14] Bae, MWTL'23
Technology	90nm SiGe	45nm CMOS SOI	130nm SiGe	65nm CMOS	65nm CMOS	65n CMOS
Design Method	End-to-end AI flow (topology, circuits, designer-input on classical/pixelated)	Manual	Manual	Automated layout & parameter optimization	Manual	Manual
Architecture	Frequency-duplexing LNA with 2-stage (Cascode, Common emitter) as decided by AI	Bi-directional PA/LNA	2-stage with feedback/compensation	2-stage CS cells	4-stage CS cells with transformers	Transformer-based band-switching load
Frequency/BW (GHz)	Combined: 24-90 Mode 1: 24-55 GHz Mode 2: 55-90 GHz	27-38	22.6-73.9	22.3-29.6/30.7	54.4-90	26-30/37-40
Gain (dB)	Mode 1: 14-17, Mode 2: 14-16	17.6	15.2	10/13.9	17.7	13.88/13.04
NF (dB)	Mode 1: 3.8-5, Mode 2: 4-6	5.2-7.8	4.06-4.94	3.8/3.47 @28GHz	5.4-7.4	3.84/4.47
DC Power (mW)	Mode 1: 24, Mode 2: 32.4	66	17.5	2.26/4.8	19	11.6
Core Area (mm ²)	0.70	0.19	0.06	0.29/0.30	0.37 (with pads)	0.17

Reference	This work: 80-160GHz Dual Peaking LNA	[15] Han, ISSCC'23	[16] Zhang, MWL'24	[17] De Filippi, JSSC'25	[18] Li, ISSCC'20	[19] Moradnia, MWTL'23
Technology	90nm SiGe	40nm CMOS	40nm CMOS	55nm SiGe	45nm CMOS SOI	90nm SiGe
Design Method	End-to-end AI flow (topology, circuits, designer-input on classical/pixelated)	Manual	Manual	Manual	Manual	Manual
Architecture	3-stage Cascode cells with RL-optimized T-lines	4-way noise-canceling combiner with CG cells	3-stage differential CS cells	3-stage Cascode cells	Noise-canceling CG/CS pair	5-stage staggered CE and Cascode cells
Frequency/BW (GHz)	75-95, 150	70-86	70-91 (S _{21,3dB})	105-175	73-88	120-157.6
Gain (dB)	30/18	16.5	16.4	23	16.8	26.5
NF (dB)	5.8 @85GHz	4.8 to 6.5	4.2-6.9	5-6.5	4.8-6.1	7.2
DC Power (mW)	63	25	20.7	60	46	20.6
Core Area (mm ²)	0.46	0.08	0.163	0.109	0.63 (with on-chip antenna)	0.69 (with pads)

Figure 13.2.7: Comparison of the LNAs with the state of the art, demonstrating new architectures, record bandwidth, and competitive performance.